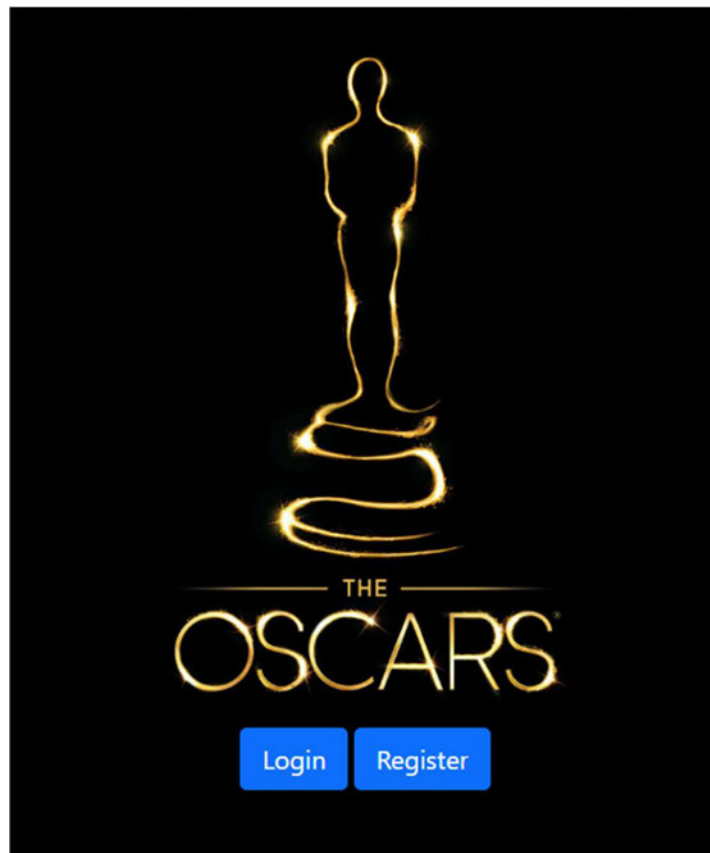


[REDACTED]

TheOscars Web Application Architecture Document

URL:

<http://theoscarsa3-env.eba-utgpapqk.us-east-1.elasticbeanstalk.com/>



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Summary

TheOscars is a web application that is designed to allow a group of people to vote on the results of The Oscars which takes place each year. The application will allow users to compare with each other if they watched a movie, what rating they gave it and a total amount of movies they have watched to see them all. Then once the ceremony has concluded an admin can update the results, which is reflected on the results page for all users to see who has won.

The application will be utilising a number of Amazon Web Services such as Elastic Beanstalk, DynamoDB, S3, EC2, API Gateway, Lambda and CloudFront. It will also utilise a third-party API OMDb allowing an admin user to retrieve Oscar movies information.

Introduction

Motivation

The motivation behind this project is from a suggestion by a friend of the game he plays with a group of his friends (I haven't participated). For the past few years it has been done based off entering in data into an excel spreadsheet (screenshots provided later) which wasn't very efficient, didn't look nice and had no way of keeping information private from each other. My friend has asked me for a few years now to see if I could do anything for him and due to my extremely limited time this has been difficult. Thankfully this project ties in well with the requirements of this assignment so this felt like a good opportunity to take his idea and see what I can do with it. Hopefully after this course I will then be able to take what I've learnt and created and move it to a personal AWS account to allow them to use it for the upcoming awards season starting in January 2023.

High-Level View

The features/functions of this will application will include the following:

1. The ability to register and create an account to login to the application.
2. An admin login will be created where an admin will be able to setup the system for all users before going live. This will require the user to search for each movie and assign it to a category. The admin will also be able to finalise the results of the ceremony for each category when announced.
3. Once logged in users will be able to:
 - a. View a dashboard where they can view each category and the associated nominated movies. Here they will be able to select that they have watched the movie, given it their own rating and if they think it will be the category winner. They will also be able to see if other viewers have watched the movie and what rating they gave it. Finally, a total count of watched movies will be shown to compared against other users as well.
 - b. View a results page that once the admin user has selected who has won for each category they will be able to see the total score and who best predicted what they were going to win.

Beneficiaries

The initial target audience will be a group of friends, which are just ordinary everyday people. I have always kept in mind the ability to scale this application beyond a small group of people. Future enhancements would include the ability to form groups of people rather than just one big group like it currently is.

Once individual groups have been implemented the application would expand to the target audience of the general public, particularly people that are interested in the Oscars Ceremony. As per (Maas, 2022) there was an estimated 16.6 million viewers that watched it live, so there is potential for this to grow given nothing like this really exists.

If that was successful and there were a number of groups using this website, I would look to monetization avenues such as ads and potentially the ability to deposit money and bet on the outcome within the group. I could then take a small percentage of the money deposited. This may then lead into a potential target audience of gamblers if I wanted to start making money from this application in the future to support future development, such as supporting other types of award ceremonies.

Therefore, future planning has been considered beyond the initial target audience of the developed application to expand further to the general public.

Related Work

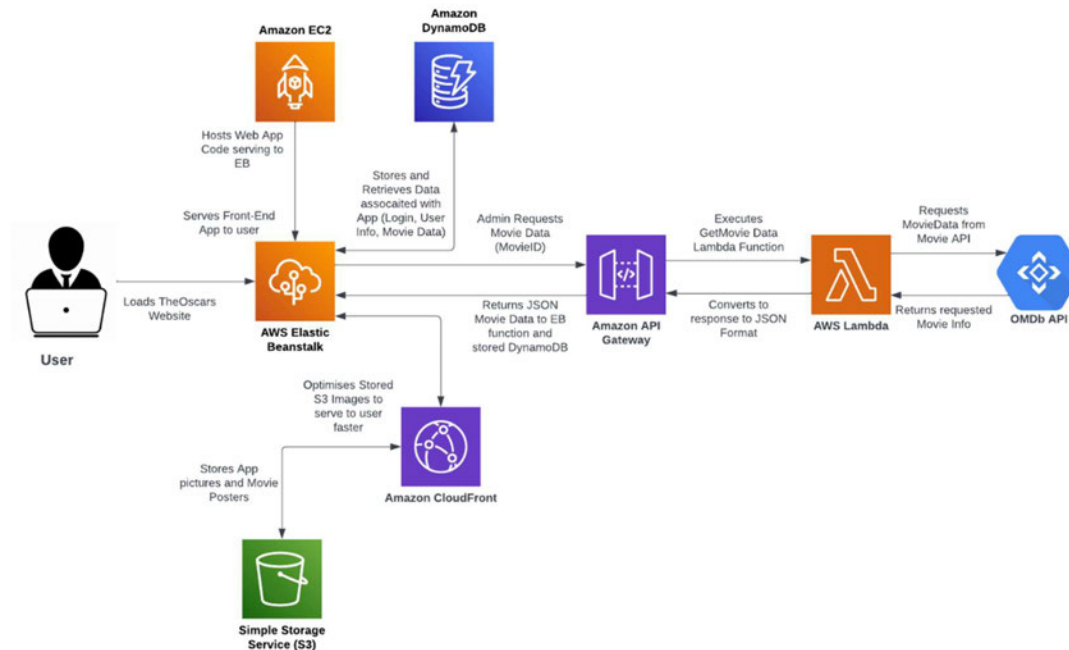
As previously mentioned this project is a brand-new idea and there isn't anything on the market that exists for something like this. I have however taken inspiration from the excel document they have been using the past few years and converted this into a web application. Screenshots as per below:

	A	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
	CATEGORIES	JOEL	MATT	ZOE	AJ	SIM	JAS	JOEL	MATT	ZOE	AJ	SIM	JAS	WINNER			
8	Best Picture							Dune	CODA	Power of the D	Power of the D	Power of the D	Power of the Dog	CODA		4. AJ & Zoe	7
9	Belfast								1							5. Jas	No score
10	CODA																
11	Don't Look Up																
12	Drive My Car																
13	Dune																
14	King Richard																
15	Licorice Pizza																
16	Nightmare Alley																
17	The Power of the Dog																
18	West Side Story																
19																	
20	Best Director							Power of the D	Power of the D	Kenneth Brana	Steven Spielberg	Steven Spielberg	WSS	Dog			
21	Kenneth Branagh - Belfast							1	1								
22	Ryusuke Hamaguchi - Drive My Car																
23	Paul Thomas Anderson - Licorice Pizza																
24	Jane Campion - The Power of the Dog																
25	Steven Spielberg - West Side Story																
26																	

	A	B	C	D	E	F	G
1							
2	#	FEATURES	MINS	NOMS	CINEMAS	STREAMING	DOWNLOAD
3	1	The Father	97	6	1 April		
4	2	Judas and the Black Messiah	126	6	Out Now		Y
5	3	Mank	131	10		Netflix	
6	4	Minari	115	6	Out Now		Y
7	5	Nomadland	107	6	Out Now		Y
8	6	Promising Young Woman	113	5	Out Now		Y
9	7	Sound of Metal	120	6		Prime	
10	8	The Trial of the Chicago 7	129	6		Netflix	
11	9	Ma Rainey's Black Bottom	94	5		Netflix	
12	10	United States vs Billie Holiday	130	1	29 April		Y
13	11	Pieces of a Woman	126	1		Netflix	
14	12	One Night in Miami	114	3		Prime	
15	13	Borat 2	95	2		Prime	
16	14	Hillbilly Elegy	116	2		Netflix	
17	15	The White Tiger	125	1		Netflix	
18	16	Da 5 Bloods	154	1		Netflix	
19	17	Eurovision	123	1		Netflix	
20	18	The Life Ahead	94	1		Netflix	
21	19	Greyhound	91	1		Apple+	
22	20	News of the World	118	4		Netflix	
23	21	Tenet	150	2			Y
24	22	Emma	124	2		Prime	
25	23	Pinocchio	125	2			Y
26	24	Mulan	115	2		Disney+	
27	25	Love and Monsters	109	1			Y
28	26	Midnight Sky	118	1		Netflix	
29	27	The One and Only Ivan	95	1		Disney+	
30		ANIMATED					

System Architecture

System Overview



System Components

The below sub-sections will describe each of the services that are used to host and run TheOscars Web application.

Elastic Beanstalk (EB)

Elastic Beanstalk is a service that is used to deploy the web applications code, which at the same time is automatically loaded into the EC2 instance that comes by default with EB. This service will take care of the infrastructure for me and provide the automatic handling of capacity provisioning, load balancing, scaling, and application health monitoring (*What is AWS Elastic Beanstalk? - AWS elastic beanstalk*). This will be particularly important as I look to further development and expand my application to the world one day, in the future for the increased website traffic.

Elastic Compute Cloud (EC2)

EC2 is an expandable virtual server that hosts and runs an instance of my web application on it. EC2 comes with EB which hosts a single instance of my application at this stage. However as mentioned above it will be able to automatically generate more virtual servers to handle the potential increased traffic to my website in the future (*What is Amazon EC2? - amazon elastic compute cloud*).

DynamoDB

DynamoDB is a NoSQL server that hosts all the data for my web application. This includes login credentials, custom user information and movie data downloaded via the Web API. I selected DynamoDB due to the flexibility and structureless requirements for my application. While a relational database may have better suited the current development and stage of my application. As I look forward to one day hopefully include TV shows, movies, and music for other award ceremonies to my application, I need the freedom of flexibility NoSQL will provide me (*What is Amazon dynamodb? - amazon dynamodb*).

Simple Storage Service (S3)

S3 is another service that is automatically generated when creating an EB service. By default, this bucket contains temporary configuration files that are required for my application to operate (*Using elastic beanstalk with Amazon S3*). I'm further utilising this bucket to store all my custom website images that I have created, along with the Movie Posters that are downloaded from the OMDb API.

CloudFront

Cloudfront is an additional complimenting web service to my S3 Bucket. By enabling and linking this service to the bucket it will speed up the distribution of all my websites content (code files and stored images). To summarise it finds the closet edge server (location) to the user and services that content directly to them. This will be an important service particularly when I launch it beyond my current region.

API Gateway

API Gateway service for my application is used for connecting to the OMBb API, which contains all the required movie data. The gateway provides a link between my web application to retrieve the POST movie data and the Lambda function which is GETTING it from the OMDb API.

Lambda

Lambda is reliant on my API Gateway as a way to communicate between it and my OMDb API. Inside here I have function that has been passed through a movie ID from my web application, through the API Gateway to the lambda function. From here it makes a request to the OMDb API for the specific movie ID and then converts it to JSON format. It is then passed back through the API Gateway and to my web application where it then selects the required data to be entered into the DynamoDB.

OMDb API

This is a paid third-party service which retrieves movie information from its database which is linked to IMDB. Here it is simply a matter of passing through any IMDB movie ID in the url through the services listed above and its return all required information for my web application (Fritz, *OMDb API*). For example, a call is made to the web url using IMDB movie tt4154796 which is for “Avengers: Endgame”. This then returns the JSON data as per the screenshot below. I then read all the required data and save it to my own database.

```
{
  "Title": "Avengers: Endgame",
  "Year": "2019",
  "Rated": "PG-13",
  "Released": "26 Apr 2019",
  "Runtime": "181 min",
  "Genre": "Action, Adventure, Drama",
  "Director": "Anthony Russo, Joe Russo",
  "Writer": "Christopher Markus, Stephen McFeely, Stan Lee",
  "Actors": "Robert Downey Jr., Chris Evans, Mark Ruffalo",
  "Plot": "After the devastating events of Avengers: Infinity War (2018), the universe is in ruins. With the help of remaining allies, the Avengers assemble once more in order to reverse Thanos' actions and restore balance to the universe.",
  "Language": "English, Japanese, Xhosa, German",
  "Country": "United States",
  "Awards": "Nominated for 1 Oscar. 70 wins & 132 nominations total",
  "Poster": "https://m.media-amazon.com/images/M/MV5BMTc5MDE2ODcwNV5BMl5BanBnXkFtZTgwMzI2NzQ2NzI@._V1_SX300.jpg",
  "Ratings": [
    {
      "Source": "Internet Movie Database",
      "Value": "8.4/10"
    },
    {
      "Source": "Rotten Tomatoes",
      "Value": "94%"
    },
    {
      "Source": "Metacritic",
      "Value": "78/100"
    }
  ],
  "Metascore": "78",
  "imdbRating": "8.4",
  "imdbVotes": "1,112,060",
  "imdbID": "tt4154796",
  "Type": "movie",
  "DVD": "30 Jul 2019",
  "BoxOffice": "$658,373,000",
  "Production": "N/A",
  "Website": "N/A",
  "Response": "True"
}
```

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